Sterling Bridge Enterprise Network

Multi-Site Enterprise Network Design | Cisco Packet Tracer (2026)

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Overview

Sterling Bridge Enterprise Network is a self-directed, end-to-end enterprise network design built entirely in Cisco Packet Tracer. It models a fictional company with three geographically distributed sites — New York, Texas, and Georgia — each supporting multiple departments, with Texas serving as the network's central hub for shared services and internet connectivity. The project was designed to mirror how a real mid-size enterprise consolidates infrastructure across locations while keeping departmental traffic logically separated, routed efficiently, and resilient to link failure.

Network Architecture

The topology spans three sites connected over a WAN backbone running OSPF, with each site built around a Layer-3 multilayer switch handling inter-VLAN routing for its local departments.

New York	Texas — Hub Site	Georgia
HR department	IT department	Sales department
Operations department	Centralized DHCP/DNS server	Marketing department
IT department	Web/FTP server	
	ISP edge & NAT/PAT gateway	

Each site uses 2960-series access switches per department feeding into a 3560-series multilayer switch, with a 2911 series router providing the WAN uplink. Texas additionally connects to a dedicated ISP-edge router (1941 series) and an external public web server used purely to validate NAT/PAT translation from outside the network.

Technical Implementation

- VLAN segmentation: each department (HR, Operations, IT, Sales, Marketing) isolated into its own VLAN across all three sites, reducing broadcast domains and enforcing logical separation of traffic.
- Inter-VLAN routing: implemented via switch virtual interfaces (SVIs) on each site's multilayer switch, avoiding router-on-a-stick bottlenecks.
- WAN routing: OSPF deployed across all three sites for dynamic, redundant site-to-site routing rather than static routes, so the network adapts automatically to topology changes.
- Centralized network services: a single DHCP/DNS server hosted at the Texas hub serves all sites via DHCP relay (ip helper-address) on remote VLANs, modeling how enterprises consolidate services instead of running per-site servers.
- Internet edge: one consolidated NAT/PAT boundary at the Texas ISP edge translates internal addressing for outbound access, tested against an external server built specifically to validate the translation.
- Documentation: full IP addressing plan, VLAN scheme, per-device configurations, and a written verification/testing plan authored alongside the build — not just the topology itself.

Testing & Validation

Beyond building the topology, the design was stress-tested: a WAN link between sites was deliberately taken down in simulation mode to confirm OSPF would automatically re-converge traffic over the remaining path without manual intervention. This validated that the redundancy built into the WAN design would hold up under a real failure scenario, not just in a static, unchallenged topology.

Skills Demonstrated

- Routing & Switching — OSPF, VLANs, inter-VLAN routing (SVI), trunking, spanning tree
- Network Services — DHCP relay, DNS, NAT/PAT, TCP/IP
- Design & Documentation — IP addressing plans, VLAN schemes, device configuration standards
- Verification & Troubleshooting — failure simulation, convergence testing, Cisco IOS CLI
- Tools — Cisco Packet Tracer, Cisco IOS